

5.5.2 Integration by Substitution (Definite Integrals)

Theorem: If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

Note: We see from the above theorem that we have to worry about changing the bounds when it comes to performing a u -substitution on definite integrals.

Example 1: Find $\int_0^{\sqrt{3}} 2x\sqrt{1+x^2}dx$.

Example 2: Find $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \cot^3(x)\csc^2(x)dx$

Example 3: Determine each of the following.

a) Find $\int_1^2 \frac{dx}{(3-5x)^2}$.

b) Find $\int_1^e \frac{\ln(x)}{x} dx$

c) Find $\int_0^1 e^{(e^x+x)} dx$

Symmetry & Integrals

Theorem (Integrals of Symmetric Functions):

Suppose f is continuous on $[-a, a]$.

- If f is even $[f(-x) = f(x)]$, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.
- If f is odd $[f(-x) = -f(x)]$, then $\int_{-a}^a f(x) dx = 0$.

General Idea: Consider the two following example functions which illustrate the above theorem.

Even: $f(x) = x^2$

Odd: $f(x) = x^3$

Proof:

Example 4: Find $\int_{-2}^2 (x^6 + 1) dx$.

Example 5: Find $\int_{-1}^1 \frac{\tan(x)}{1 + x^2 + x^4} dx$.